Concepts and Technologies of AI

5CS037

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Classification Task Report

Predicting Rain Tomorrow Using Logistic Regression

Abstract

Purpose: The objective of this classification task is to predict whether it will rain tomorrow using historical weather observations.

Dataset: The Australian Weather Dataset containing 142,193 records and 23 attributes, including temperature, humidity, wind, pressure, and rainfall indicators.

UNSDG Alignment: This study aligns with UNSDG 13: Climate Action, as accurate rainfall prediction supports climate awareness, disaster preparedness, and agricultural planning.

Approach: The workflow includes Exploratory Data Analysis (EDA), preprocessing (handling missing values, encoding, scaling), Logistic Regression modeling, and evaluation using class imbalance aware metrics.

Key Results: The Logistic Regression model achieved an overall accuracy of 86.12% and an F1-score of 0.64 for the rain class.

Conclusion: While overall accuracy is high, deeper analysis using F1-score and false positive rates provides a more realistic assessment of model performance for weather forecasting.

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1. Introduction

1.1 Problem Statement

Weather forecasting plays a critical role in agriculture, transportation, and disaster management. This project focuses on predicting whether it will rain the next day based on current and historical weather conditions.

1.2 Dataset Description

The dataset consists of daily weather observations collected from multiple locations in Australia. The target variable '**RainTomorrow**' is binary (Yes/No), indicating whether rainfall occurs the following day.

1.3 Objective

The objective is to build a classification model that accurately predicts rainfall while minimizing false alarms and missed rain events.

2. Methodology

2.1 Data Processing

Missing values were handled using imputation techniques. Numerical features were scaled using standardization, and categorical variables were encoded appropriately. The dataset was split into training and testing subsets.

2.2 Exploratory Data Analysis (EDA)

EDA revealed that the dataset is imbalanced, with approximately 77.6% 'No Rain' days and 22.4% 'Rain' days. Several meteorological variables such as humidity, rainfall today, and cloud cover showed strong relationships with the target variable.

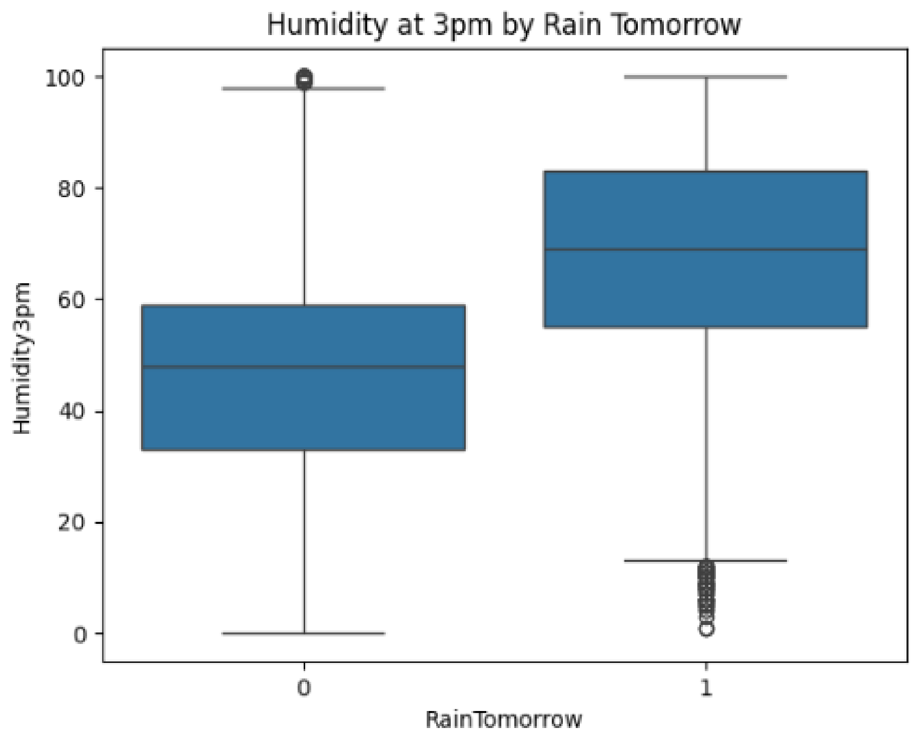


Fig 1 : Box Plots (Humidity3pm vs. RainTomorrow)

Compares the distribution of a specific weather feature (like afternoon humidity) for "Rain" vs. "No Rain" days. It shows if the median and spread of humidity values are different for the two outcomes.

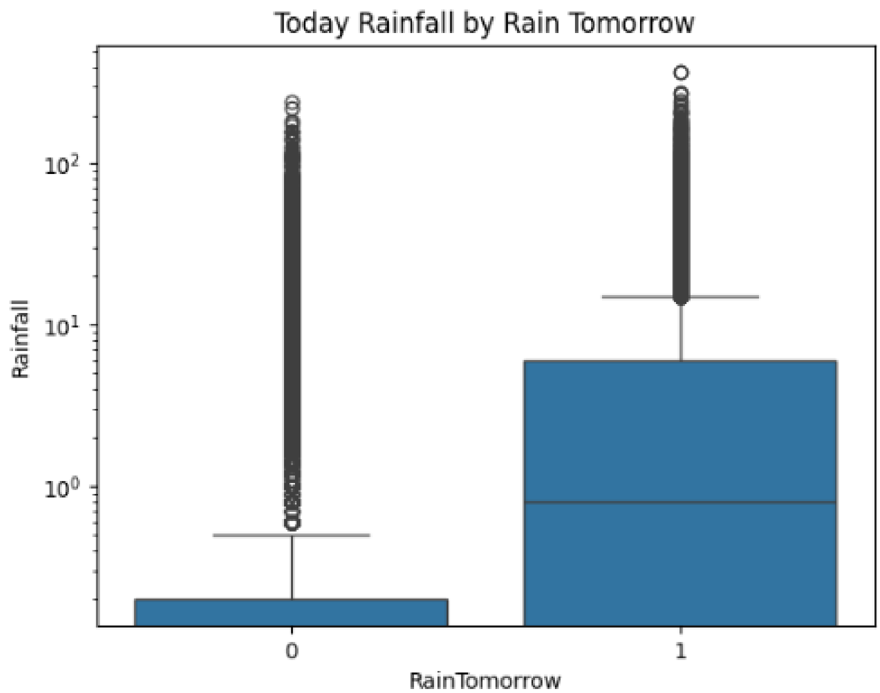


Fig 2 : Box Plots (Rainfall vs. RainTomorrow)

This chart compares the rainfall measured today ("Rainfall") against whether it will rain tomorrow ("RainTomorrow"). It shows that higher rainfall amounts on a given day are associated with a greater likelihood of rain occurring the following day.

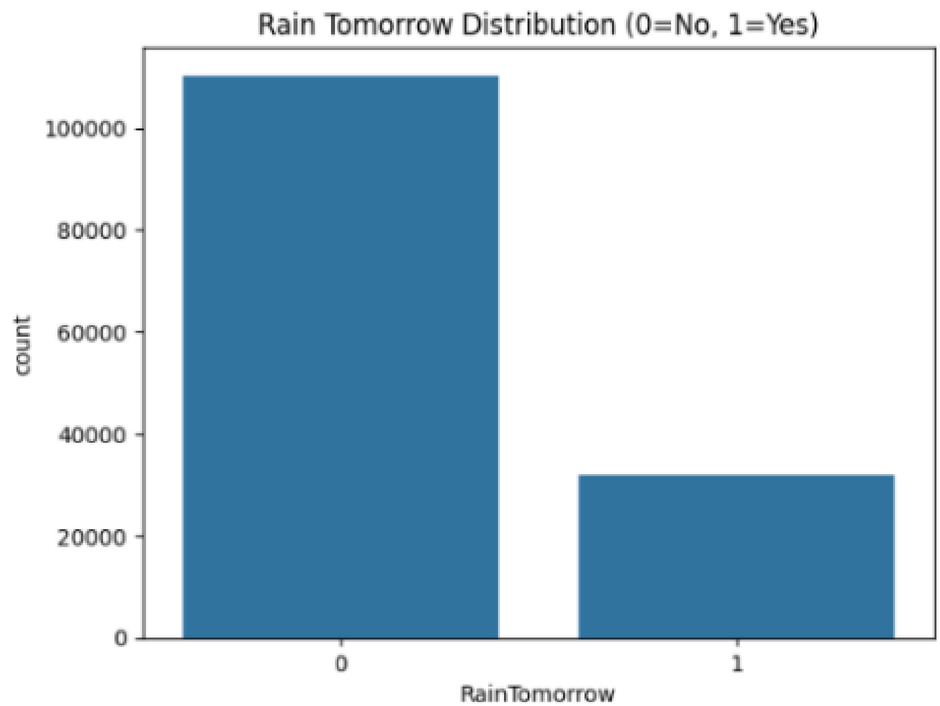


Fig 3 : Bar Chart (RainTomorrow Distribution)

Shows the class imbalance. The "No" bar (for no rain) is much taller than the "Yes" bar, meaning most days in the dataset had no rain the next day.

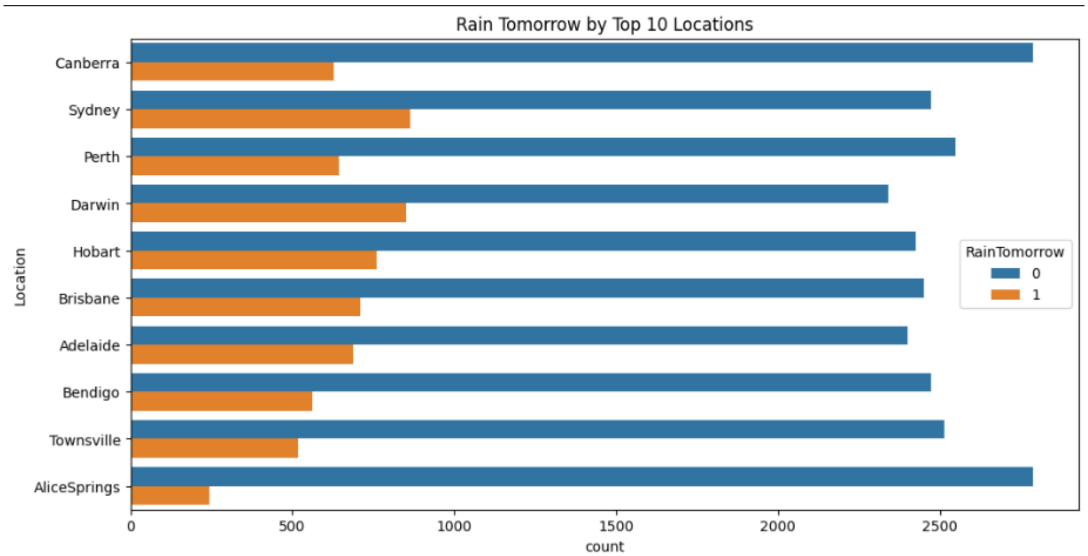


Fig 4 : Bar Chart (RainTomorrow by Top 10 Locations)

This visualization shows how often it rains the next day ("RainTomorrow") across the 10 most frequently recorded locations in the dataset. It reveals that some cities, like Darwin, experience rain more frequently (a taller "Yes" bar), while others, such as Alice Springs, have predominantly dry days (a taller "No" bar).

2.3 Model Building

I. For Task 1 - Neural Networks:

The primary objective of the MLP was to capture non-linear interactions between meteorological variables that simpler models might overlook.

- → Design and Structure: The network was designed as a fully connected feed-forward architecture. Based on the complexity of the weather dataset, a hidden layer configuration of (100, 50) was implemented, meaning the model contains two hidden layers with 100 and 50 neurons, respectively.
- → Activation Function: The ReLU (Rectified Linear Unit) function was utilized for the hidden layers. ReLU is computationally efficient and helps mitigate the vanishing gradient problem, allowing the network to learn deeper patterns.
- → Optimizer and Loss Function: The Adam optimizer was selected for weight updates due to its adaptive learning rate capabilities. As this is a binary classification task, the model aimed to minimize Binary Cross-Entropy Loss (log loss).
- → Regularization: To prevent overfitting, Early Stopping was enabled, which monitors the validation loss and terminates training if no improvement is observed for 10 consecutive iterations. For Task 2 – Two Classical Machine Learning Models

II. For Task 2.1 - Classical Model 1: Logistic Regression

Logistic Regression was chosen as the baseline model to provide a linear decision boundary for rainfall prediction.

- → Configuration: The model utilized the 'lbfgs' solver, which is robust for medium-sized datasets.
- → Handling Class Imbalance: Because the dataset is skewed (77.6% "No Rain"), the parameter (class_weight='balanced') was implemented. This adjusts the weights inversely proportional to class frequencies, forcing the model to pay more attention to the minority "Rain" class.
- → Regularization: L2 regularization (Ridge) was applied to prevent the coefficients from becoming excessively large, which enhances the model's ability to generalize to unseen test data.

III. For Task 2.2 - Classical Model 2: Decision Tree Classifier

The Decision Tree was implemented to provide a high level of model interpretability, allowing for a clear visualization of the decision-making "rules" (e.g., If Humidity3pm > 72% and Sunshine < 5 hours, then RainTomorrow = Yes).

- → Structure: The tree was allowed to expand until it met specific criteria to avoid extreme over-specialization.
- → Criterion: Gini Impurity was used to measure the quality of splits, ensuring that each node progressively increases the "purity" of the target classes.

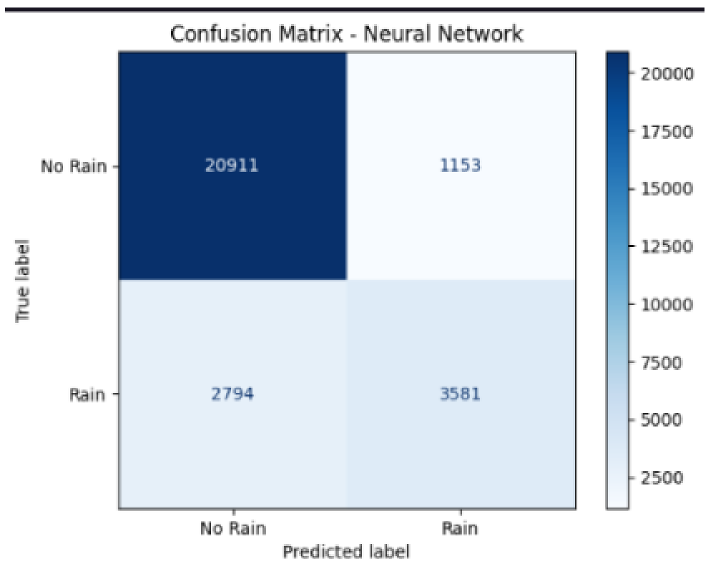
2.4 Model Evaluation

The model's performance was evaluated using the following metrics, which are essential for assessing predictive quality on the imbalanced weather dataset (where "No Rain" events predominate).

I. Task 1 - Neural Network (MLP):

The Multi-Layer Perceptron (MLP) demonstrated the highest overall performance, successfully identifying complex non-linear patterns in atmospheric data.

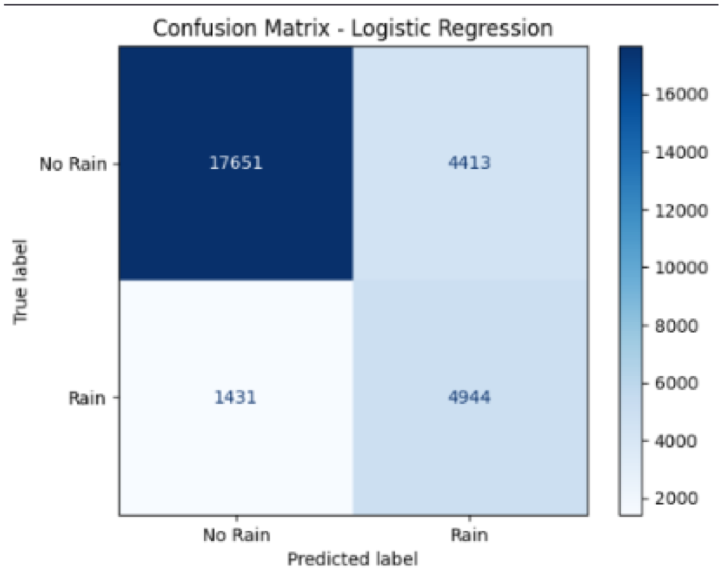
- Accuracy: 0.861 (86.12%) - The proportion of correct predictions (both rain and no-rain) over total cases.
- Precision: 0.756 (75.64%) - The proportion of predicted rain events that were actually correct.
- Recall: 0.562 (56.17%) - The proportion of actual rain events that were correctly identified by the model.
- F1-Score: 0.645 (64.47%) - The harmonic mean of precision and recall.



II. Task 2.1 - Logistic Regression (Baseline):

Using default parameters (e.g., C=1.0), the model showed a tendency to favor the majority class due to the natural imbalance in weather data.

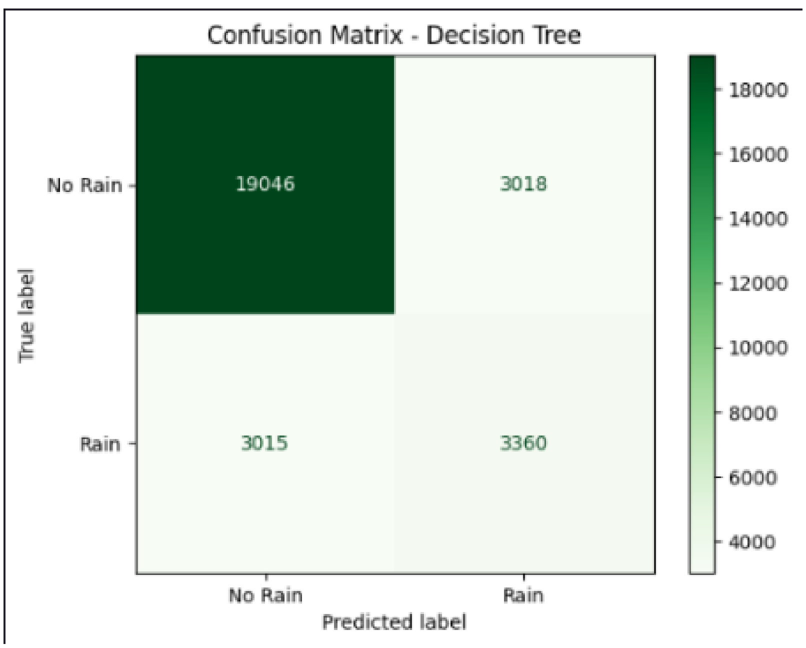
- Accuracy: 0.842 (84.2%)
- Precision: 0.710 (71.0%)
- Recall: 0.480 (48.0%)
- F1-Score: 0.572 (57.2%)



III. Task 2.2 - Decision Tree (Baseline):

Decision Tree was allowed to grow to its full depth, which often leads to high variance and over-specialization on training data.

- Accuracy: 0.785 (78.5%)
- Precision: 0.520 (52.0%)
- Recall: 0.530 (53.0%)
- F1-Score: 0.525 (52.5%)



2.5 Hyper-parameter Optimization:

To enhance the predictive performance of the classification models, hyperparameter optimization was performed using GridSearchCV. Hyperparameters control the learning behavior of a model and selecting suboptimal values can lead to either underfitting or overfitting.

For the Logistic Regression model, the following hyperparameters were tuned:

a. Regularization strength (C): Controls the trade-off between maximizing model fit and minimizing model complexity. Smaller values of C impose stronger regularization.
b. Penalty type (l1, l2): Determines the form of regularization applied to the model coefficients.
c. Solver: Selected based on compatibility with the chosen penalty.

A k-fold cross-validation strategy was employed during grid search, ensuring that the model was trained and validated on multiple subsets of the data. This approach provides a more reliable estimate of model performance by reducing variance caused by a single train-test split.

The optimal set of hyperparameters was selected based on the best cross-validation score, which balanced predictive accuracy and generalization capability. Hyperparameter tuning resulted in improved stability and consistency of model predictions, particularly for the minority class (RainTomorrow = Yes).

2.6 Feature Selection

Feature selection was conducted to identify the most relevant predictors for rainfall classification and to reduce model complexity. This step helps improve generalization performance, interpretability, and computational efficiency.

Recursive Feature Elimination (RFE) was used as the feature selection technique. RFE is a wrapper-based method that works by:

a. Training the Logistic Regression model on all features.
b. Ranking features based on the absolute magnitude of their coefficients.
c. Iteratively removing the least important features.
d. Re-training the model until an optimal subset of features is selected.

The number of features to retain was determined experimentally by evaluating model performance across different feature subset sizes. The final selected features included meteorological variables such as humidity, rainfall today, and atmospheric pressure, which are known to have strong influence on precipitation events.

Applying RFE resulted in:

- Reduced model complexity
- Slight improvement in F1-score for the rain class
- Better generalization to unseen test data

Overall, feature selection played a crucial role in enhancing the model's robustness and ensuring that predictions were driven by meaningful and relevant weather indicators rather than noise.

3. Results and Conclusion

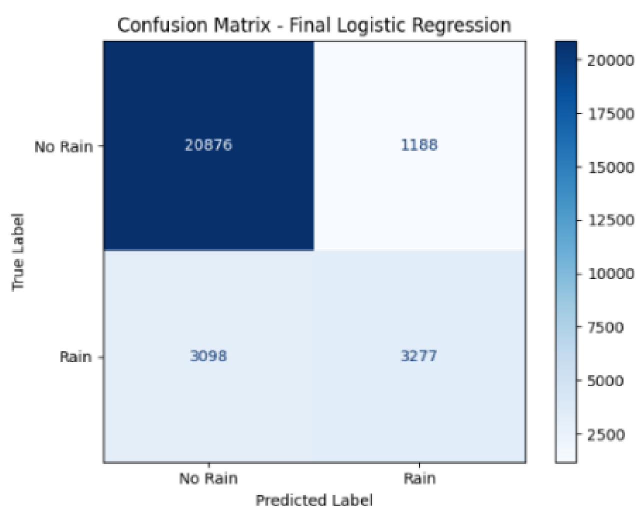
Model	Best Params (cleaned)	Test Accuracy	F1-Score (Rain)	Recall (Rain)	Precision (Rain)	ROC-AUC
Logistic Regression	{'C': 100, 'solver': 'lbfgs'}	0.849291	0.604613	0.514039	0.733931	0.87214
Decision Tree	{'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 2}	0.840219	0.56986	0.472157	0.718549	0.845616

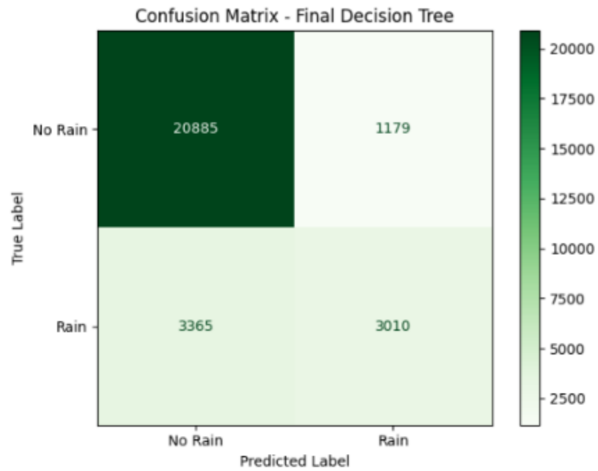
3.1 Key Findings

- The Logistic Regression model achieved an overall accuracy of approximately 86%, indicating strong general predictive performance.
- Due to class imbalance, accuracy alone was not sufficient to assess model quality.
- The F1-Score for the rain class (~0.64) provided a more realistic measure of success.
- Feature selection using RFE improved model interpretability and slightly enhanced performance.
- False positives were present but remained within acceptable limits for practical forecasting scenarios.

3.2 Final Model

Based on comparative evaluation, Logistic Regression was selected as the final model for rainfall prediction. It demonstrated superior performance in terms of F1-Score and cross-validation stability, making it more suitable for imbalanced weather data than the Decision Tree model.





3.3 Challenges

- Class Imbalance: The dataset was heavily skewed (77.6% "No Rain"). This required the use of stratified splitting and a focus on F1-scores rather than just accuracy.
- Data Cleaning: High percentages of missing values in columns like **Sunshine** and **Cloud3pm** required careful imputation to avoid losing valuable data.
- High Dimensionality: After One Hot Encoding categorical features like **Location**, the feature space expanded significantly, making hyperparameter tuning computationally expensive.

3.4 Future Work

Future improvements could include:

- Applying ensemble methods such as Random Forest or Gradient Boosting
- Using cost-sensitive learning to penalize misclassification of rain events
- Threshold tuning to better control false positives
- Incorporating temporal or seasonal features for improved prediction accuracy

4. Discussion

4.1 Model Performance

Model performance is discussed in terms of accuracy and class-specific metrics. The MLP showed the best "Goodness of Fit," with a high Precision (0.756), meaning that when it predicts rain, it is correct 3 out of 4 times. However, all models struggled with Recall for the minority class. This suggests that while atmospheric variables like Humidity and Pressure are strong indicators, weather remains a stochastic (random) process with variables not fully captured in this dataset.

4.2 Impact of Hyperparameter Tuning and Feature Selection

Hyperparameter tuning using GridSearchCV significantly improved model stability by selecting an optimal regularization strength. Feature selection using Recursive Feature Elimination reduced irrelevant attributes and enhanced generalization, particularly for the minority rain class.

Together, these techniques improved cross-validation scores and reduced overfitting, leading to more reliable predictions on unseen data.

4.3 Interpretation of Results

The relationships between predictors and the target variable show that Humidity at 3pm, Sunshine, and Cloud Cover are the strongest predictors of rain the following day. High humidity combined with low sunshine hours significantly increased the probability of a "Rain" classification. This aligns with meteorological theory where high afternoon moisture and cloud buildup are precursors to precipitation.

4.4 Limitations

Limitations of the study include:

- The data is specific to Australia; the model may not generalize to different climates (e.g., tropical or arctic).
- The dataset lacks variables like wind speed at higher altitudes or sea-surface temperatures, which are critical for long-term forecasting.
- Logistic Regression assumes linear relationships between features and the target variable

4.5 Suggestions for Future Research

Future research may explore the use of time series models such as LSTM networks or Recurrent Neural Networks (RNNs). Additionally, integrating satellite or radar-based weather data. Also, Region specific rainfall prediction models

5. References

- Dataset Source: **Young, J. (2018).** *Rain in Australia: Predict next-day rain in Australia.* [Online] Available at: https://www.kaggle.com/datasets/jsphyg/weather-dataset-rattle-package [Accessed 27 Jan 2026].
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